POLYSENSE PROTOCOL

Common frame structure

Field	Size (bytes)	Type	Notes
SYNC	2	const	0x55, 0xAA (not included in CRC)
ver	1	uint8	Protocol version (start at 0x01)
msg_type	1	uint8	See message type table below
node_id	1	uint8	1–255 (0 = broadcast/reserved)
len	2	uint16LE	Payload length in bytes
payload	len	bytes	Message-type specific
crc16	2	uint16LE	CRC-16/CCITT over ver..last payload byte

Message types

Name	ID (hex)	Direction	Purpose
ENV_TELEM	0x01	Node → GS	Environment (T/P/H) + timing
IMU_TELEM	0x02	Node → GS	Accel/gyro + optional quaternion
GPS_TELEM	0x03	Node → GS	GPS position/time/fix quality
STATUS	0x04	Node → GS	Battery, storage, mission/link status
BEACON	0x05	Node → GS	Compact recovery frame
CMD	0x10	GS → Node	Commands (recovery, motor, mission, cfg)
CMD_ACK	0x11	Node → GS	Command acknowledgments

ENV_TELEM (0x01)

Field	Size	Type	Units / Meaning
uptime_ms	4	uint32	Node uptime in ms
sample_id	4	uint32	Incrementing per ENV_TELEM
temperature_c_x100	2	int16	°C × 100
pressure_pa	4	uint32	Pascals
humidity_rh_x100	2	uint16	%RH × 100; 65535 if invalid

Field	Size	Type	Units / Meaning
env_flags	2	uint16	Bitfield (validity, sensor type)
reserved	8	bytes	All zero now (future env data)

env_flags bits

Bit(s)	Meaning
0	Temperature valid
1	Pressure valid
2	Humidity valid
8–15	Env sensor type (0=BME280,1=BME688, …)

IMU_TELEM (0x02)

Field	Size	Type	Units / Meaning
uptime_ms	4	uint32	Node uptime in ms
sample_id	4	uint32	Incrementing per IMU_TELEM
ax_mps2_x1000	2	int16	m/s² × 1000
ay_mps2_x1000	2	int16	m/s² × 1000
az_mps2_x1000	2	int16	m/s² × 1000
gx_dps_x1000	2	int16	deg/s × 1000
gy_dps_x1000	2	int16	deg/s × 1000
gz_dps_x1000	2	int16	deg/s × 1000
imu_flags	2	uint16	Validity, sensor type, etc.
qw_x10000	2	int16	Quaternion w × 10000 (0 if unused)
qx_x10000	2	int16	Quaternion x × 10000
qy_x10000	2	int16	Quaternion y × 10000
qz_x10000	2	int16	Quaternion z × 10000
reserved	8	bytes	Future (e.g. mag, accel bias)

imu_flags bits

Bit(s)	Meaning
0	Accel valid
1	Gyro valid
2	Quaternion valid
8–15	IMU type (0=MPU6500,1=ICM42688, …)

GPS_TELEM (0x03)

Field	Size	Type	Units / Meaning
uptime_ms	4	uint32	Node uptime in ms
sample_id	4	uint32	Incrementing per GPS_TELEM
lat_deg_x1e7	4	int32	Latitude (deg × 1e7)
lon_deg_x1e7	4	int32	Longitude (deg × 1e7)
alt_m_x100	4	int32	Altitude (m × 100)
ground_speed_cmps	2	uint16	Ground speed (cm/s)
course_deg_x100	2	uint16	Track over ground (deg × 100)
hdop_x100	2	uint16	HDOP × 100
sat_count	1	uint8	Satellites used in fix
fix_type	1	uint8	0=no fix, 2=2D, 3=3D, etc.
gps_flags	2	uint16	Extra flags (constellations, etc.)

STATUS (0x04)

Field	Size	Type	Units / Meaning
uptime_ms	4	uint32	Node uptime in ms
battery_mv	2	uint16	Battery voltage (mV)
battery_pct_x100	2	uint16	% × 100, 65535 if unknown
storage_free_kb	4	uint32	Free storage (kB), 0xFFFFFFFF if N/A
mission_state	1	uint8	Mission phase enum
recovery_flags	1	uint8	Beacon, low battery, GPS lost, etc.
link_rssi_dbm	2	int16	Link RSSI (dBm), 0x7FFF if N/A
link_snr_db_x10	2	int16	SNR × 10, 0x7FFF if N/A
status_flags	2	uint16	Misc flags
reserved	2	bytes	Future

mission_state

Value	State
0	IDLE
1	PRELAUNCH
2	ASCENT
3	DESCENT
4	LANDED
5	RECOVERY

BEACON (0x05)

Field	Size	Type	Units / Meaning
uptime_ms	4	uint32	Node uptime in ms
last_fix_age_ms	2	uint16	Age of last GPS fix (ms), 0xFFFF if N/A
lat_deg_x1e7	4	int32	Last known latitude
lon_deg_x1e7	4	int32	Last known longitude
alt_m_x100	4	int32	Last known altitude (m × 100)
battery_mv	2	uint16	Battery voltage
recovery_flags	1	uint8	As in STATUS
beacon_seq	1	uint8	Incrementing beacon sequence
reserved	4	bytes	Future

CMD (0x10)

Common command payload header (first 6 bytes):

Field	Size	Type	Meaning
cmd_id	2	uint16	Unique ID per command from GS
cmd_class	1	uint8	0=RECOVERY,1=MOTOR,2=MISSION,3=CONFIG
cmd_code	1	uint8	Specific command within class
param_len	2	uint16	Length of following params in bytes
params	N	bytes	Binary parameters per command type

Command classes

Class	Value	Meaning
RECOVERY	0	Recovery aids
MOTOR	1	Motor control
MISSION	2	Mission state
CONFIG	3	Configuration

Command codes and param layouts

1. RECOVERY_SET_BEACON (class 0, code 0x01)

Field	Size	Type	Meaning
enable	1	uint8	0=off, 1=on
period_ms	4	uint32	Beacon interval in ms

2. MOTOR_SET_SIDES (class 1, code 0x01)

Field	Size	Type	Meaning
left_speed_x1000	2	int16	Normalized [-1.0,1.0] × 1000
right_speed_x1000	2	int16	Same
flags	1	uint8	Bits for direction/brake/mode
reserved	1	uint8	Future

3. MOTOR_STOP_ALL (class 1, code 0x02)

Field	Size	Meaning
(none)	0	param_len=0

4. MISSION_SET_STATE (class 2, code 0x01)

Field	Size	Type	Meaning
mission_state	1	uint8	As in STATUS

5. MISSION_SET_RATES (class 2, code 0x02)

Field	Size	Type	Meaning
env_hz_x100	2	uint16	ENV_TELEM rate (Hz × 100)
imu_hz_x100	2	uint16	IMU_TELEM rate (Hz × 100)
gps_hz_x100	2	uint16	GPS_TELEM rate (Hz × 100)
status_hz_x100	2	uint16	STATUS rate (Hz × 100)

CMD_ACK (0x11)

Field	Size	Type	Meaning
cmd_id	2	uint16	Echo of CMD's cmd_id
status	1	uint8	Result code
detail	1	uint8	Optional detail / error code
timestamp_ms	4	uint32	Uptime when command processed

Status codes

Value	Meaning
0	RECEIVED (queued)
1	ACCEPTED
2	DONE

Value	Meaning
3	REJECTED_INVALID
4	REJECTED_STATE
5	ERROR_INTERNAL